Correction notice

Genomic Identity of the Sima de los Huesos Hominins: Convergent Evidence from Mitochondrial and Nuclear DNA — version 2

Emilio García-Morán · 21 October 2026 · Supersedes version 1 (doi:10.5281/zenodo.19389351, April 2026) and the October 2026 manuscript *Phylogenetic placement of Sima de los Huesos hominins revisited*.

Summary

Part	Status
Mitochondrial DNA	**Corrected.** With explicit *Pan troglodytes* rooting, Sima is sister to Denisova, as reported by Meyer et al. (2014). The “basal Sima” placement is withdrawn.
Description of Meyer et al. (2014)	**Corrected.** They used a Bayesian analysis (MrBayes, GTR+I+Γ; ~120 mtDNAs including 46 chimpanzee and bonobo genomes; D-loop removed), not an unrooted Neighbor-Joining tree.
Nuclear DNA (D-statistics, distances)	**Under verification.** Withdrawn as results until the original ANGSD outputs are recovered and checked.
Phase 2 mtDNA analyses	**Withdrawn.** Input data included non-hominin sequences.

1. Mitochondrial DNA

What was claimed. Version 1 described Sima de los Huesos as the basal sister lineage to (Denisova + Neandertal), with 93% IQ-TREE bootstrap support, and interpreted the Sima–Denisova affinity reported by Meyer et al. (2014) as an artefact of an unrooted analysis.

What went wrong. The IQ-TREE run was performed without specifying the outgroup; its own output states that the tree is unrooted and that Sima is only *drawn* at the root. The 93% value supports the split {Sima, Denisova} | {Neandertals, *Pan*}, which, rooted on *Pan*, is the Sima + Denisova clade. Reading Sima as basal was a misreading of an unrooted tree, and the published tree figures were typed by hand rather than drawn from the output. The original computations were correct: the March 2026 RAxML-NG bootstrap replicates already supported Sima + Denisova in 98.4%.

Corrected result. Six complete mitochondrial genomes (Sima Femur XIII KF683087, Denisova 3, Vindija 33.16, Feldhofer 1, Mezmaiskaya 1, *Pan troglodytes*), re-analysed from GenBank with automatic validation of every record, MAFFT 7.526, IQ-TREE 3.0.1 and RAxML-NG 1.2.2, all rooted on *Pan*. Bootstrap support (% of 1,000 replicates):

Alignment	Analysis	Sima + Denisova	Sima + Neandertals	Denisova + Neandertals
full	IQTREE TPM2u+F+G4 UFBoot	**97.9**	2.1	0.0
full	IQTREE HKY+G4 standard bootstrap	**92.9**	7.1	0.0
full	RAxML GTR+G standard bootstrap	**97.9**	2.1	0.0
no D-loop	IQTREE HKY+F+G4 UFBoot	**97.2**	0.9	1.9
no D-loop	IQTREE HKY+G4 standard bootstrap	**98.8**	0.4	0.8
no D-loop	RAxML GTR+G standard bootstrap	**99.6**	0.3	0.1

Joint topology tests (10,000 RELL replicates) reject “Sima basal” in all four model/alignment combinations; Sima + Denisova has the highest likelihood in all four:

Alignment	Model	Hypothesis	ΔlogL	p-AU
full	HKY+G4	H0 Sima basal (withdrawn claim)	10.52	**0.014**
full	HKY+G4	H1 Sima + Denisova	0.00	0.948
full	HKY+G4	H2 Sima + Neandertals	8.74	0.067
full	TPM2u+F+G4	H0 Sima basal (withdrawn claim)	12.35	**0.006**
full	TPM2u+F+G4	H1 Sima + Denisova	0.00	0.968
full	TPM2u+F+G4	H2 Sima + Neandertals	10.68	**0.042**
no D-loop	HKY+G4	H0 Sima basal (withdrawn claim)	6.39	**0.043**
no D-loop	HKY+G4	H1 Sima + Denisova	0.00	0.959
no D-loop	HKY+G4	H2 Sima + Neandertals	6.39	**0.042**
no D-loop	HKY+F+G4	H0 Sima basal (withdrawn claim)	6.39	**0.043**
no D-loop	HKY+F+G4	H1 Sima + Denisova	0.00	0.959
no D-loop	HKY+F+G4	H2 Sima + Neandertals	6.39	**0.042**

Figure 1. Rooted mitochondrial phylogeny: Sima de los Huesos is sister to Denisova

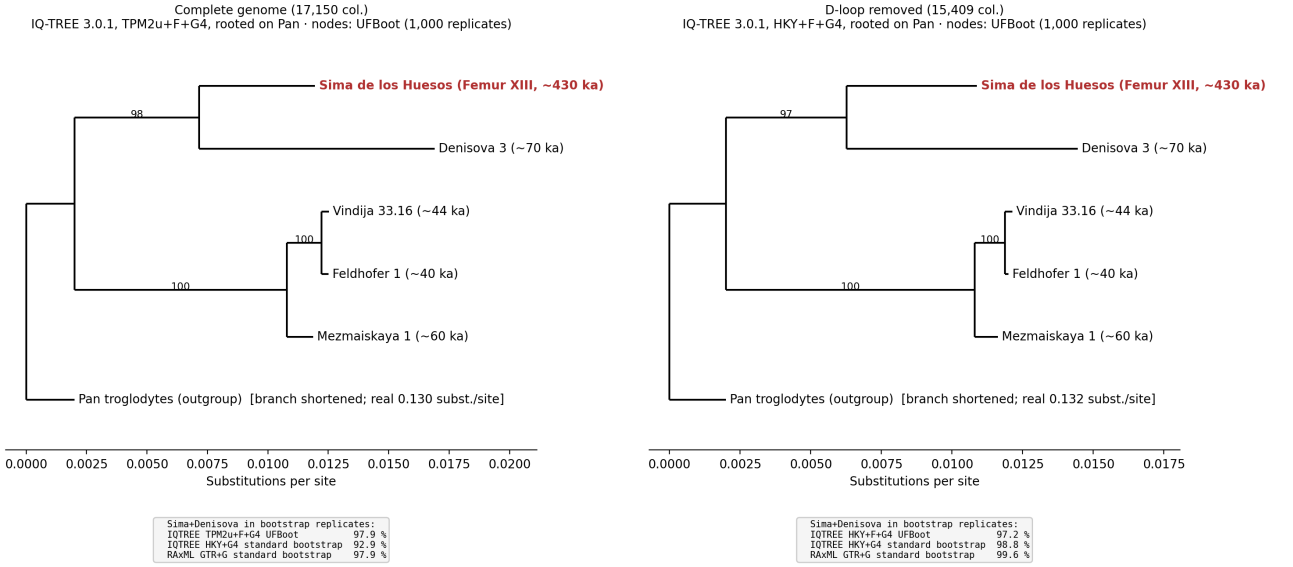


Figure 1. Rooted mitochondrial phylogeny drawn directly from the IQ-TREE output (UFBoot values at nodes; *Pan* branch shortened for display, real length indicated).

This agrees with Meyer et al. (2014) and with Posth et al. (2017), who also rooted on chimpanzee. The mito-nuclear discordance is explained by the replacement model: Sima retains the ancestral, Denisovan-like mtDNA of the early Neandertal lineage, later replaced in Neandertals by an introgressed African mtDNA between ~413 and ~268 ka (Posth et al. 2017); a parallel replacement is documented for the Y chromosome (Petr et al. 2020).

2. Phase 2 analyses (withdrawn)

Accession numbers in the phase 2 scripts were typed by hand and never checked against the records GenBank returned. Ten are not hominin mitochondrial DNA:

Used as	Accession	Actual GenBank record
Oase 1, Tianyuan, Kostenki 14, Vindija 33, Denisova 4/8	KX198073/74/75/78/79	*Agaricus* (fungus), EF1α gene
“Sima SH5”	KY751401	*Henneguya tucunarei* (fish parasite), 18S rRNA
Ust’-Ishim	KP295291	*Phytophthora castaneae* (oomycete), HSP90
Ust’-Ishim	KC758896	*Ostrinia scapulalis* (moth), ABC transporter
“Denisova 2”	NC_024829	*Nephroselmis astigmatica* (green alga), chloroplast genome
“Neandertal”	EU260658	Simian immunodeficiency virus, partial *tat* gene

Phase 2 results, including the “6–0” Sima–Denisova shared-site count, are withdrawn. The six genomes of the corrected analysis are unaffected; their GenBank titles and sequence hashes are now validated before alignment.

3. Nuclear DNA (under verification)

The nuclear section of version 1 (chromosome 1 only; D = 0.7425, Z = 30.6; pairwise distances) cannot yet be reproduced: the ANGSD commands and outputs were not archived with the code, and the figure scripts contain hand-typed values. These numbers are withdrawn as results while the original outputs are recovered. A genome-wide, fully reproducible nuclear analysis is planned as a separate project. The taxonomic proposal of version 1 is withdrawn with it.

4. Data, code and reproducibility

Repository: https://github.com/EmilioGarciaMoran/sima-genomics (CORRECTION.md; analysis in mtDNA_repro/). One command, bash mtDNA_repro/run_all.sh genbank, reproduces the analysis from GenBank to the figure (~16 min on 2 cores). Repeated runs give identical results. UFBoot and the topology tests run single-threaded, since multithreaded their values shifted slightly with the thread count (never changing a conclusion); with that setting, two different machines (2 and 4 threads, sequences downloaded from GenBank) produced byte-identical result tables and this PDF. Superseded scripts and manuscripts remain in the repository with notices. This PDF is generated by mtDNA_repro/scripts/07_correction_notice.py directly from the result files.

References

Meyer M. *et al.* (2014) A mitochondrial genome sequence of a hominin from Sima de los Huesos. *Nature* 505:403–406. doi:10.1038/nature12788

Meyer M. *et al.* (2016) Nuclear DNA sequences from the Middle Pleistocene Sima de los Huesos hominins. *Nature* 531:504–507. doi:10.1038/nature17405

Posth C. *et al.* (2017) Deeply divergent archaic mitochondrial genome provides lower time boundary for African gene flow into Neanderthals. *Nat. Commun.* 8:16046. doi:10.1038/ncomms16046

Petr M. *et al.* (2020) The evolutionary history of Neanderthal and Denisovan Y chromosomes. *Science* 369:1653–1656. doi:10.1126/science.abb6460